Mathematical Analysis Fixture

数学分析回归样本：本页验证中英文正文、标题层级与行内公式。

For a sequence (a_n), convergence to L means every epsilon admits a tail bound.

$$\lim_{n \to \infty} a_n = L$$

Theorem and Proof

Theorem

If f is differentiable at x_0, then f is continuous at x_0.

Proof

Write $f(x)-f(x_0)$ as a difference quotient times $x-x_0$ and take the limit.

1. Isolate the difference quotient.
2. Apply the product limit law.

$$f(x)-f(x_0)=\frac{f(x)-f(x_0)}{x-x_0}(x-x_0)$$

Table and Figure

The table records a convergent sequence.

n	1	2	4	8
1/n	1	0.5	0.25	0.125

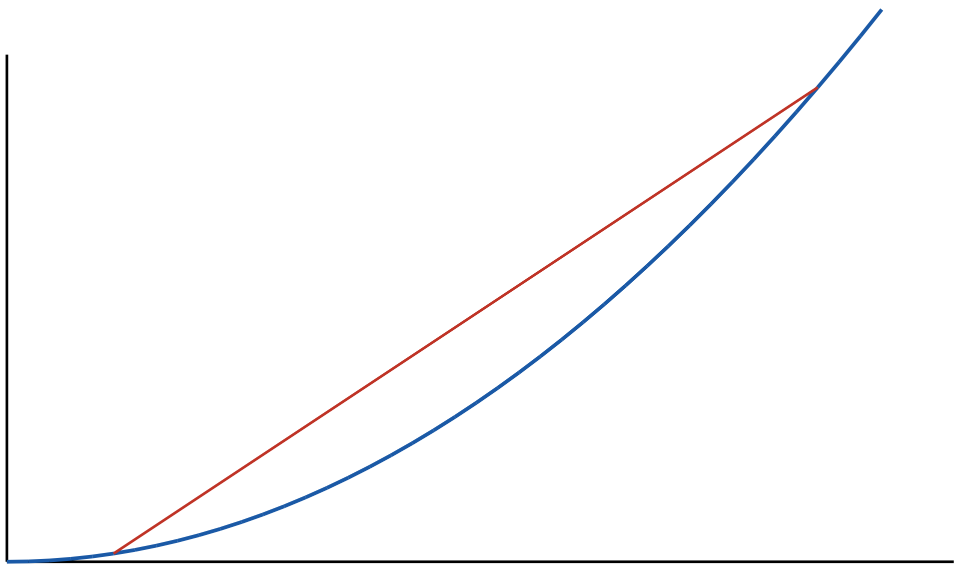

Figure 1. Secant slopes approaching the tangent line.

Two-column Reading Order

Left column

Definition. A set is open when every point has a contained neighborhood.

Example. Every open interval (a,b) is open in the real line.

Right column

Definition. A set is closed when it contains all of its limit points.

Observation. Complements exchange open and closed sets.